■ Mathematics Ka Mukammal Safar

Counting se PhD tak — Umra k Mutabiq Poora Syllabus

Ye document ek maa-baap ya khud seekhne wale bacche ke liye tayyar kiya gaya hai. Isme mathematics ke **har topic ko umra ke mutabiq** tarteeb diya gaya hai — 3 saal ki simple counting se lekar PhD level ke advanced mathematics tak. Har topic ke saath approximate umra di gayi hai jab tak woh topic seekh lena chahiye. Ye ek guide hai, strict rule nahi — har bachche ki apni raftaar hoti hai.

PHASE	AGE	LEVEL
Phase 1	3–5 Saal	Pre-School / Nursery
Phase 2	6–8 Saal	Primary (Class 1–3)
Phase 3	9–11 Saal	Middle Primary (Class 4–6)
Phase 4	12–14 Saal	Upper Middle (Class 7–9)
Phase 5	15–17 Saal	Secondary / High School
Phase 6	18–21 Saal	Undergraduate / BSc
Phase 7	22–25 Saal	Postgraduate / MSc
Phase 8	25+ Saal	Research / PhD Level

PHASE 1: Pre-School Foundation (Umra: 3–5 Saal)

3–4 Saal: Ginti Ki Bunyaad

- 1 se 10 tak ginti karna (bolna aur sunna)
- Ungliyon se ginti karna
- Cheezein ginni — apples, blocks, toys
- Bohot, Thora, Zyada, Kam — concept samajhna
- Bada / Chhota pehchanna (size comparison)
- Shapes pehchanna: Gol, Teen kona, Chaukona, Aayat
- Rangon se sorting karna
- Pattern banana: ABAB wala sequence (laal-neela-laal-neela)
- Upar / Neche / Aage / Peechhe — position words

4–5 Saal: Numbers aur Shapes Ka Gehrayee

- 1 se 20 tak ginti aur likhna
- Number recognition (1–20)
- Joda (pair) aur Tanha (single) ka farq
- Sorting: Akar, Rang, Shape ke hisaab se
- Simple jorna: 1+1=2, 2+2=4 (objects se)
- Simple ghatana: 3-1=2 (objects hata ke dikhana)
- Zyada / Kam comparison (kaun zyada hai?)
- 2D shapes: Circle, Triangle, Square, Rectangle, Hexagon
- 3D shapes: Sphere (gend), Cube (dibba), Cylinder
- Time ka basic concept: Subah, Dopahar, Shaam, Raat
- Measurement: Lamba / Chhota, Bhari / Halki
- Patterns: AABB, ABC ABC
- Ordinal numbers: Pehla, Doosra, Teesra

■ *Parents ke liye Note: Is umra mein khel khel mein sikhao. Blocks, kandil, ginti wali games use karo.*

PHASE 2: Primary Mathematics (Umra: 6–8 Saal, Class 1–3)

6 Saal (Class 1): Numbers aur Basic Operations

- 1 se 100 tak ginti, likhna, padhna
- Ones, Tens (Ikayi, Dasai) — place value
- Jorna (Addition) bina carry ke: 23+14
- Ghatana (Subtraction) bina borrow ke: 35-12
- Even aur Odd numbers (Jift aur Taaq)
- Skip counting: 2s, 5s, 10s mein
- Shapes ki sides aur angles ginni
- Paisa: 1, 2, 5, 10 rupee pehchanna
- Time: Ghanta aur minute (clock padhna)
- Data: Simple tally marks
- Measurement: Ruler se lamba maapna (cm)

7 Saal (Class 2): Operations ka Vistar

- 1 se 1000 tak numbers
- Place value: Hundreds, Tens, Ones
- Addition with carrying (haath aana wali jama)
- Subtraction with borrowing (haath maangna)
- Multiplication tables: 1 se 5 tak
- Division ka concept: barabar baantna
- Half (adha), Quarter (chauthai) — fractions ka taaruf
- Calendars: Din, Hafta, Mahina, Saal
- Weight: Grams aur Kilograms
- Capacity: Litre aur Millilitre
- Perimeter ka basic concept (line se gher ke maapna)
- Graphs: Bar graph banana aur padhna
- 3D shapes: Faces, Edges, Vertices ginni

8 Saal (Class 3): Multiplication aur Fractions

- Multiplication tables 1 se 10 tak (yaad karna zaroori)
- Long multiplication (2-digit x 1-digit)
- Division: Short division method
- Remainders samajhna
- Fractions: 1/2, 1/3, 1/4, 2/3 — picture se samajhna
- Comparing fractions (kaun bada?)
- Decimals ka pehla taaruf: .5 = adha
- Roman numerals (I, V, X, L, C)
- Area ka concept (squares ginke)
- Perimeter calculations
- Angles: Right angle pehchanna
- Line of symmetry
- Word problems (masail)

- Mental math tricks

■ *Note: Multiplication tables yaad hona bohot zaroori hai. Roz 10 minute practice karo.*

PHASE 3: Middle Primary (Umra: 9–11 Saal, Class 4–6)

9 Saal (Class 4): Number System Aur Geometry

- 5-digit aur 6-digit numbers
- Lakhs aur Crores ka system
- Long multiplication: 3-digit x 2-digit
- Long division: 2-digit divisor
- Factors aur Multiples
- Prime numbers aur Composite numbers
- LCM (Laghu Samapavartya) aur HCF (Maha Samapavartya)
- Fractions: Add, Subtract (same denominator)
- Mixed numbers: 2½ samajhna
- Decimals: Add aur Subtract
- Angles: Acute, Obtuse, Right, Straight, Reflex
- Triangles ke types: Equilateral, Isosceles, Scalene
- Quadrilaterals: Square, Rectangle, Rhombus, Parallelogram
- Area of Rectangle aur Square
- Line graphs padhna
- Average (mean) calculate karna

10 Saal (Class 5): Fractions, Decimals, Percentages

- Fractions: Add aur Subtract (alag denominator)
- Fractions multiply karna
- Decimals: Multiply aur Divide
- Percentage ka concept aur calculations
- Ratio aur Proportion ka taaruf
- Unitary Method (ek ki qeemat se hisaab)
- Negative numbers ka concept (thermometer, debt)
- Integers ki line (number line)
- 3D shapes: Cube, Cuboid ka volume
- Circle: Radius, Diameter, Circumference
- Pie chart banana aur padhna
- Basic probability: Sikka uchhaalna
- Algebraic thinking: ■ + 3 = 7 type masail
- Coordinate grid ka taaruf (x, y)

11 Saal (Class 6): Pre-Algebra aur Advanced Arithmetic

- Integers: Add, Subtract, Multiply, Divide (negative bhi)
- BODMAS / PEMDAS rule
- Algebraic expressions: x+3, 2y-5
- Variables aur constants
- Simple equations: x+5=12, 2x=8
- Ratio aur Proportion — advanced
- Percentage: Discount, Profit, Loss
- Simple Interest

- Speed, Distance, Time formula
- Perimeter aur Area — Triangle, Parallelogram
- Circle: Area aur Circumference (pi ka taaruf)
- 3D: Surface area ka basic taaruf
- Data: Mean, Median, Mode, Range
- Probability: 0 se 1 tak scale
- Basic set theory: Union, Intersection (Venn diagram)
- Exponents ka taaruf: $2^3 = 8$

PHASE 4: Upper Middle School (Umra: 12–14 Saal, Class 7–9)

12 Saal (Class 7): Algebra Ka Aaghaz

- Linear equations (ek variable): $3x - 7 = 14$
- Linear equations (do variable): simultaneous equations
- Inequalities: $x > 5$, $2x + 1 \leq 9$
- Rational numbers (kasrat wale numbers)
- Irrational numbers ka concept ($\sqrt{2}$ rational nahi)
- Exponents aur Powers — laws: $a^m \times a^n = a^{(m+n)}$
- Scientific notation: 6.4×10^3
- Proportionality: Direct aur Inverse
- Percentage applications: Tax, Discount, Markup
- Compound Interest
- Geometry: Parallel lines aur transversal
- Congruent triangles (SAS, ASA, SSS)
- Pythagoras Theorem ka taaruf
- Circle theorems basics
- Coordinate geometry: Plotting points, quadrants
- Bar charts, histograms, frequency tables
- Probability: Theoretical vs Experimental

13 Saal (Class 8): Polynomials aur Geometry

- Polynomials: Add, Subtract, Multiply
- Factoring: Common factor, $a^2-b^2 = (a+b)(a-b)$
- Factoring: $x^2 + bx + c$ form
- Quadratic equations: $ax^2 + bx + c = 0$ ka taaruf
- Simultaneous linear equations (elimination, substitution)
- Graphing linear equations (slope-intercept form)
- Slope (dhalaan) aur y-intercept
- Functions ka taaruf: $f(x)$ notation
- Pythagoras Theorem — applications
- Trigonometry ka taaruf: Sin, Cos, Tan
- Similar triangles
- Area aur Volume: Cylinder, Cone, Sphere
- Surface area calculations
- Statistics: Grouped data, cumulative frequency
- Box plots (box-and-whisker)
- Scatter plots aur correlation
- Sets: Subset, Complement, De Morgan's Laws

14 Saal (Class 9): Intermediate Algebra aur Trigonometry

- Quadratic equations: Factoring, Completing the square
- Quadratic formula: $x = (-b \pm \sqrt{(b^2-4ac)}) / 2a$
- Discriminant aur nature of roots

- Quadratic graphs (parabolas)
- Inequalities: Quadratic aur absolute value
- Indices/Exponents — negative aur fractional: a^(-n), a^(1/2)
- Surds (irrational numbers): $\sqrt{2}$, $\sqrt{3}$ simplify karna
- Logarithms ka taaruf: log■■, ln
- Trigonometry: Sin, Cos, Tan angles nikalna
- Pythagorean identities: $\sin^2\theta + \cos^2\theta = 1$
- Bearing problems (navigation)
- Loci aur Constructions
- Circle theorems: Angle at centre, Tangent
- Vectors ka taaruf
- Sequences: Arithmetic aur Geometric
- nth term formula
- Permutations aur Combinations ka taaruf
- Probability: Combined events, Tree diagrams
- Standard Deviation

PHASE 5: Secondary / High School (Umra: 15–17 Saal, Class 10–12)

15 Saal (Class 10): Advanced Algebra aur Functions

- Functions: Domain, Range, Composition, Inverse
- Polynomial functions aur graphs
- Rational functions aur asymptotes
- Exponential functions: $y = a^x$
- Logarithmic functions aur laws
- Solving log equations
- Trigonometric functions: Unit circle
- Radians ka concept
- Graphs of sin, cos, tan
- Amplitude, Period, Phase shift
- Trigonometric equations solve karna
- Coordinate Geometry: Distance, Midpoint, Gradient
- Equation of a line aur circle
- Matrices: Add, Subtract, Multiply
- Determinant aur Inverse of 2×2 matrix
- Systems of equations using matrices
- Arithmetic Progression (AP): Sum formula
- Geometric Progression (GP): Sum formula
- Infinite GP: Sum to infinity
- Binomial Theorem
- Complex numbers ka taaruf: $i = \sqrt{-1}$
- Argand diagram

16 Saal (Class 11): Pre-Calculus aur Further Mathematics

- Limits ka concept (intuitive samajhna)
- Differentiation ka taaruf (rate of change)
- Basic derivatives: $d/dx(x^n) = nx^{(n-1)}$
- Derivatives: sin, cos, e^x, ln(x)
- Chain rule, Product rule, Quotient rule
- Applications: Tangent/Normal to a curve
- Maxima aur Minima (optimization)
- Integration ka taaruf (reverse of differentiation)
- Indefinite integrals
- Definite integrals aur area under curve
- Differential equations ka pehla taaruf
- Vectors: 2D aur 3D, Dot product, Cross product
- 3D Coordinate Geometry
- Proof by Mathematical Induction
- Further Trigonometry: Addition formulae
- Half angle, Double angle formulae
- Inverse trig functions: arcsin, arccos, arctan

- Complex numbers: Polar form, De Moivre's Theorem
- Number Theory basics: Modular arithmetic
- Combinatorics: nCr aur nPr advanced
- Probability Distributions: Binomial, Normal
- Hypothesis testing ka taaruf
- Linear Programming
- Graph Theory basics

17 Saal (Class 12 / A-Level): Calculus aur Advanced Topics

- Calculus: Further differentiation techniques
- Implicit differentiation
- Parametric equations ka differentiation
- Integration: Substitution method
- Integration by parts
- Partial fractions aur integration
- Differential equations: Separable variables
- First order ODEs
- Numerical methods: Newton-Raphson
- Trapezium rule aur Simpson's rule
- Further Sequences aur Series: Power series
- Taylor aur Maclaurin series ka taaruf
- Further Matrices: Eigenvalues, Eigenvectors
- Transformations with matrices
- Further Vectors: Planes, Lines in 3D
- Polar Coordinates
- Further Statistics: Correlation coefficient
- Chi-squared test
- Poisson Distribution
- Mechanics: Kinematics, Forces, Moments
- Proof by Contradiction aur Contrapositive
- Further Complex Numbers
- Hyperbolic functions: sinh, cosh, tanh

PHASE 6: Undergraduate / BSc Level (Umra: 18–21 Saal)

Year 1 (18–19 Saal): Foundation of Higher Mathematics

- Real Analysis: Sequences, Limits, Continuity ka rigorous proof
- Epsilon-Delta definition of limits
- Completeness of real numbers
- Calculus I: Riemann integration
- Fundamental Theorem of Calculus (proof ke saath)
- Multivariable Calculus: Partial derivatives
- Gradient, Divergence, Curl
- Multiple integrals: Double, Triple integrals
- Linear Algebra I: Vector spaces, Subspaces
- Linear transformations aur their matrices
- Basis aur Dimension
- Eigenvalues aur Eigenvectors
- Diagonalization
- Abstract Algebra I: Groups
- Subgroups, Cyclic groups, Lagrange's theorem
- Rings ka taaruf
- Discrete Mathematics: Logic, Set theory, Proofs
- Relations aur Functions
- Graph theory
- Number Theory: Divisibility, Primes, GCD
- Modular arithmetic, Fermat's Little Theorem
- ODE (Ordinary Differential Equations) — full course
- Complex Analysis ka taaruf
- Probability Theory I: Axioms, Random variables
- Discrete aur Continuous distributions
- Statistics: Estimation, Confidence intervals

Year 2 (19–20 Saal): Core Advanced Mathematics

- Real Analysis II: Uniform convergence
- Series of functions: Power series
- Metric spaces
- Complex Analysis: Analytic functions, Cauchy-Riemann
- Contour integration, Cauchy's integral theorem
- Residue theorem
- Linear Algebra II: Inner product spaces
- Spectral theorem, SVD
- Orthogonality, Gram-Schmidt process
- Abstract Algebra II: Rings — Ideals, Quotient rings
- Fields aur Field extensions
- Group homomorphisms, Isomorphism theorems
- Topology I: Open/closed sets, Compactness, Connectedness
- Partial Differential Equations (PDEs) I

- Heat, Wave, Laplace equations
- Fourier Series aur Transforms
- Numerical Analysis: Error analysis, Root finding
- Numerical integration, ODE numerical methods
- Differential Geometry I: Curves aur Surfaces
- Curvature
- Probability Theory II: Law of Large Numbers
- Central Limit Theorem
- Statistical inference, Regression
- Combinatorics: Generating functions
- Ramsey theory ka taaruf
- Mathematical Logic: Propositional, First-order logic

Year 3 (20–21 Saal): Specialization Courses

- Measure Theory: Lebesgue integration
- Sigma-algebras, Measurable functions
- Functional Analysis I: Normed spaces, Banach spaces
- Hilbert spaces, Bounded operators
- Algebraic Topology: Fundamental group
- Homotopy aur Homology (taaruf)
- Galois Theory
- Commutative Algebra
- Differential Geometry II: Riemannian geometry taaruf
- Manifolds
- PDE II: Sobolev spaces, Weak solutions
- Fourier Analysis advanced
- Stochastic Processes: Markov chains, Brownian motion
- Mathematical Statistics: Maximum likelihood
- Bayesian statistics
- Game Theory: Nash equilibrium
- Optimization: Convex optimization, Lagrange multipliers
- Dynamical Systems: Phase portraits, Stability
- Chaos theory
- Number Theory II: Quadratic reciprocity
- p-adic numbers ka taaruf
- Representation Theory of Groups (taaruf)
- Category Theory (taaruf)
- Mathematical Physics: Classical mechanics, QM math
- Cryptography: RSA, Elliptic curves

PHASE 7: Postgraduate / MSc Level (Umra: 22–25 Saal)

MSc Year 1–2 (22–24 Saal): Advanced Specialization

- Functional Analysis II: Spectral theory
- Operator algebras: C*-algebras
- Advanced Complex Analysis: Several complex variables
- Riemann surfaces
- Algebraic Geometry: Affine aur Projective varieties
- Schemes (Grothendieck) — taaruf
- Algebraic Topology: Singular homology, Cohomology
- Homotopy theory: Higher homotopy groups
- Fiber bundles
- Differential Geometry III: Connection, Curvature forms
- Characteristic classes
- Lie Groups aur Lie Algebras
- Representation Theory: Advanced
- Homological Algebra: Exact sequences, Derived functors
- Category Theory: Functors, Natural transformations, Adjunctions
- Topos theory (taaruf)
- Advanced Number Theory: Algebraic number theory
- L-functions aur modular forms
- Advanced PDEs: Microlocal analysis
- Elliptic operators
- Advanced Probability: Stochastic calculus (Ito's lemma)
- Stochastic differential equations
- Mathematical Finance: Black-Scholes
- Advanced Dynamical Systems: Ergodic theory
- Symplectic Geometry
- Model Theory (Mathematical Logic)
- Set Theory: Cardinals, Ordinals, Axiom of Choice
- Forcing aur Independence results (taaruf)
- Computability Theory: Turing machines, Decidability
- Complexity Theory: P vs NP

PHASE 8: Research / PhD Level (Umra: 25+ Saal)

Research Level: Frontier Mathematics

Ye wo topics hain jo Harvard, MIT, Princeton PhD programs mein cover hote hain:

■ Algebraic Geometry

— Schemes aur Morphisms
— Sheaves aur Cohomology
— Derived categories
— Moduli spaces
— Intersection theory
— Mirror Symmetry
— Langlands Program

■ Number Theory

— Elliptic curves aur Wiles' theorem (Fermat's Last)
— Automorphic forms aur representations
— Arithmetic geometry
— p-adic L-functions
— Iwasawa theory
— The Langlands correspondence
— Analytic number theory: Distribution of primes

■ Topology aur Geometry

— 4-Manifold theory (Donaldson, Seiberg-Witten)
— 3-Manifolds aur Geometrization (Perelman)
— Symplectic aur Contact topology
— Floer homology
— Higher category theory (∞-categories)
— Motivic cohomology
— K-theory

■ Analysis aur PDEs

— Harmonic Analysis: Calderón-Zygmund theory
— Dispersive PDEs (Schrödinger, Wave equations)
— Geometric measure theory
— Free boundary problems
— Navier-Stokes regularity (Millennium Problem)
— Spectral geometry
— Non-linear functional analysis

■ Probability aur Mathematical Physics

— Conformal field theory
— Random matrices
— Integrable systems
— Statistical mechanics: Phase transitions
— Quantum groups
— String theory mathematics
— Topological quantum field theory

■ Logic aur Foundations

— Large cardinal axioms
— Descriptive set theory
— Proof theory: Cut elimination
— Homotopy type theory (HoTT)
— Univalent foundations
— Categorical logic
— Reverse mathematics

■ Combinatorics aur Discrete Math

— Extremal combinatorics
— Algebraic combinatorics
— Additive combinatorics (Green-Tao theorem)
— Probabilistic method
— Spectral graph theory
— Ramsey theory (advanced)
— Topological combinatorics

■ Millennium Prize Problems

— P vs NP Problem — Complexity Theory
— Navier-Stokes Existence aur Smoothness
— Riemann Hypothesis
— Yang-Mills Existence aur Mass Gap
— Hodge Conjecture
— Birch aur Swinnerton-Dyer Conjecture

■ PhD mein sirf padhai nahi, RESEARCH hoti hai — matlab naya mathematics create karna. Apne supervisor ke saath kisi specific area mein original contribution karna hota hai.

Mukhtasir Khulasa — Umra se Syllabus

Umra	Class/Level	Mukhya Topics
3–4 Saal	Pre-Nursery	Ginti 1-10, Shapes, Rang, Patterns
4–5 Saal	Nursery/KG	Ginti 1-20, Basic jorna/ghatana, Time
6 Saal	Class 1	100 tak numbers, Place value, Jorna/Ghatana
7 Saal	Class 2	1000 tak, Tables 1-5, Fractions taaruf
8 Saal	Class 3	Tables 1-10, Decimals, Fractions, Area
9 Saal	Class 4	LCM/HCF, Fractions operations, Angles
10 Saal	Class 5	Percentage, Ratio, Integers, Algebra taaruf
11 Saal	Class 6	Equations, BODMAS, PI, Statistics
12 Saal	Class 7	Linear Equations, Pythagoras, Trig taaruf
13 Saal	Class 8	Polynomials, Quadratics, Geometry, Scatter
14 Saal	Class 9	Quadratic Formula, Logs, Sequences, SD
15 Saal	Class 10	Functions, Matrices, Trig advanced, Complex
16 Saal	Class 11	Calculus taaruf, Vectors, Proof, Stats
17 Saal	Class 12	Full Calculus, ODEs, Further Algebra
18–21 Saal	BSc	Real Analysis, Algebra, Topology, Measure
22–24 Saal	MSc	Algebraic Geometry, Lie Groups, Stochastic
25+ Saal	PhD	Research, Millennium Problems, Original Work

Ye syllabus NCERT, Cambridge IGCSE/A-Level, MIT OpenCourseWare, aur Harvard Mathematics curriculum ke mutabiq tayyar kiya gaya hai. Doosri PDF mein in sab topics ke liye free online resources, YouTube channels, aur practice question links diye gaye hain.